Physics Part 3

Periodic Motion

Periodic motion is a type of motion that regularly repeats, meaning the object returns to its initial position after some time or simply motions that repeats on a regular cycle.

A special type of vibratory or periodic motion is called **simple harmonic motion**. Simple harmonic motion is a motion that occurs when the restoring force on an object is directly proportional to the object's displacement from equilibrium. For simple harmonic motion, the acceleration of the object is proportional to its position and is oppositely directed to the displacement from equilibrium.

$$a = -\frac{k}{m}x$$

An example of a system that follows simple harmonic motion is the motion of a block on a frictionless surface connected to a spring.

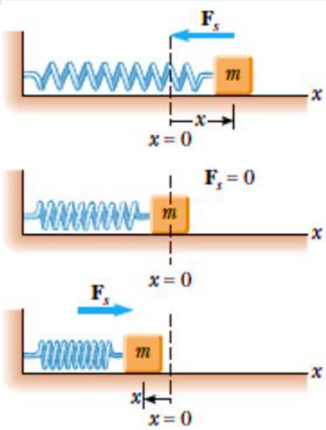


Figure shows a Spring and Block system of Simple Harmonic Motion

Period (T) is the time required for the system to proceed through one complete cycle.

$$T = 2\pi\sqrt{\frac{m}{k}}$$

m – mass of the block (kg)
k – spring constant / force constant (N/m)

The value of the **spring constant**, k, can be derived using Hooke's Law:

$$F = kx$$

F – force applied to the spring (N)
x – displacement of the spring (m)

The **frequency** of the motion indicates how frequently the motion repeats itself. It has a unit Hertz, named after the German scientist, Heinrich Hertz.

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

m – mass of the block (kg)
k – spring constant / force constant (N/m)

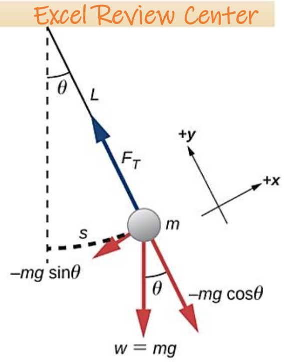
The frequency can also be expressed in terms of angular frequency:

$$\omega = 2\pi f = \sqrt{\frac{k}{m}}$$

m – mass of the block (kg)
k – spring constant / force constant (N/m)

Simple Pendulum

Another system that follows simple harmonic motion is a **simple pendulum**. A simple pendulum consists of a concentrated bob, with mass m, attached to a string, with length L, of negligible mass.



The period, the time required for the system to proceed through one complete cycle, is:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where:
L – length of the pendulum's string (m)
g – acceleration due to gravity (9.81 m/s²)

The angular frequency of the pendulum is:

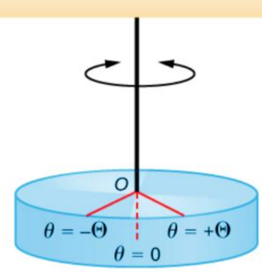
$$\omega = 2\pi f = \sqrt{\frac{g}{L}}$$

where:
f – frequency of oscillation (Hz)
L – length of the pendulum's string (m)
g – acceleration due to gravity (9.81 m/s²)

Notice that the period, frequency and angular frequency of the pendulum is independent of the mass of the bob.

Torsional Pendulum

Another system that follows simple harmonic motion is a **torsional pendulum**. This type of pendulum twists back and forth with respect to its equilibrium position.



The period, the time required for the system to proceed through one complete cycle, is:

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

where:
I – moment of inertia
κ – torsional constant

Waves

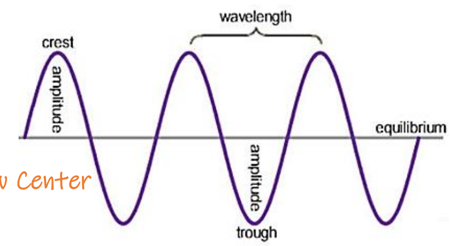
A **wave** is defined as a repeating and periodic disturbance that travels through a medium. A medium is any substance or material that carries the wave. As a wave travels through a certain medium, energy is being transported from one point to another with transporting matter.

Mechanical Waves

Mechanical waves are waves that requires certain medium in order for them to be able to transport energy. Two types of mechanical waves are transverse waves and longitudinal waves.

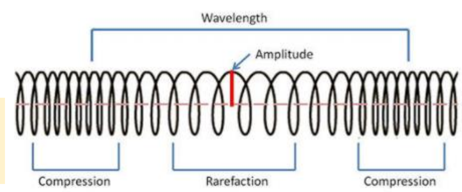
Transverse Wave

When the paths in which the particles of the medium vibrate are **perpendicular** to the motion of the wave, it is said to be a **transverse wave**.



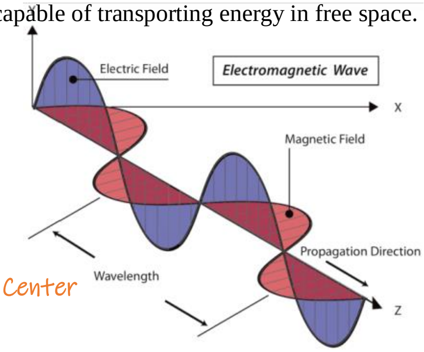
Longitudinal Wave

When the paths in which the particles of the medium vibrate are **parallel** to the motion of the wave, it is said to be a **longitudinal wave**.



Electromagnetic Wave

Electromagnetic waves are waves that are capable of transporting energy in free space.



Speed of wave

Transverse Wave

$$v = \sqrt{\frac{T}{\mu}}$$

where:
T - tension the string is stretched (N)
μ - mass per unit length of string (kg/m)

Longitudinal wave

$$v = \sqrt{\frac{E}{\rho}}$$

where:
E - modulus of elasticity (N/m²)
ρ - density (kg/m³)